

Unit 11

Writing Statistical Conclusions

Lecture notes for IB DP Biology and Chemistry

Course purpose	To help students move from raw experimental data to cautious, justified scientific claims.
Unit focus	Result paragraphs, claim strength, effect size, uncertainty, p-values, graphs, limitations, and IB-style evaluation language.
Central habit	Do not write that a hypothesis was proved. State what the data suggest, how large the effect was, how uncertain it remains, and what limits the claim.

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1 Purpose of Unit 11

By this point, students have met uncertainty, distributions, sample size, p-values, t-tests, ANOVA, chi-square tests, regression, and diagnostics. The remaining problem is not calculation. It is writing.

A student can choose the right test and still write a weak conclusion. This usually happens when the student jumps from a numerical result to an oversized claim.

Core thesis

A statistical conclusion is a controlled scientific claim. It must connect the pattern in the data, the size of the effect, the uncertainty, the statistical evidence, and the limitations of the design.

By the end of this unit, students should be able to:

1. distinguish a data description from a scientific conclusion;
2. write a result paragraph using effect size, uncertainty, and statistical evidence;
3. avoid common invalid phrases such as “proved”, “no effect”, and “human error”;
4. report non-significant results without claiming the null hypothesis is true;
5. connect graphs, p-values, and biological or chemical meaning;
6. write limitations as mechanisms, not vague excuses;
7. produce IB-style conclusion and evaluation paragraphs.

2 Data description versus scientific claim

A data description says what was observed. A scientific claim says what the observation supports, given uncertainty and design limitations.

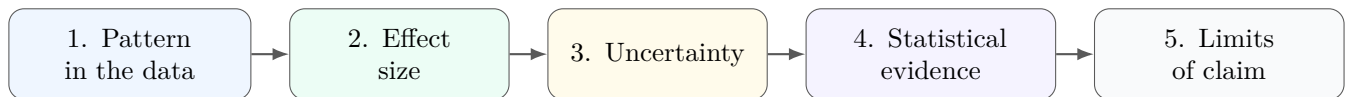
Statement type	Example
Raw observation	The treatment group had taller plants than the control group in the sample.
Processed result	The treatment mean was 2.8 cm higher than the control mean.
Statistical evidence	A Welch t-test gave $p = 0.018$, suggesting the observed difference would be unusual under equal population means.
Scientific claim	The data suggest that the fertilizer increased plant height under the tested conditions, although the conclusion is limited by the small sample size and uncontrolled light variation.

Common error

Do not stop at “the graph shows a difference”. A graph can show a pattern, but the conclusion must explain whether the pattern is large, uncertain, statistically supported, and scientifically meaningful.

3 The claim pipeline

A strong statistical conclusion moves through a fixed sequence.



One-sentence version

The result should say: what changed, by how much, how uncertain that estimate is, how the statistical test behaved, and what the result does and does not justify.

A useful formula for students is:

Conclusion = pattern + effect size + uncertainty + evidence + caution.

4 A paragraph anatomy

A full result paragraph does not need to be long. It needs to be complete.

Example paragraph: plant growth

The fertilizer group had a higher mean plant height than the control group after 14 days. The observed mean difference was 2.8 cm, with the treatment group showing both higher central tendency and some overlap with the control group. A Welch t-test indicated that this difference would be unlikely under the null hypothesis of equal means, $p = 0.018$. These data support the claim that fertilizer increased plant height under the tested conditions, but the conclusion is limited by the small sample size and possible variation in light exposure between trays.

Part	What it does
Pattern	States the direction of the observed result.
Effect size	Says how large the difference or relationship was.
Uncertainty	Signals variation, overlap, confidence interval, or sample-size limits.
Statistical evidence	Reports the p-value or interval without overinterpreting it.
Scientific caution	Restricts the claim to the actual design and conditions.

5 Bad phrases and better replacements

Many weak IA conclusions use the same phrases. They sound scientific but are too strong, too vague, or logically wrong.

Weak phrase	Better version
The hypothesis was proved.	The data support the hypothesis under the tested conditions.
The hypothesis was correct.	The observed pattern was consistent with the prediction.
There was no effect.	No statistically significant effect was detected.
The result was due to human error.	Endpoint judgement may have introduced variation in titration volume.
The graph shows a relationship.	The scatter plot suggests a positive relationship, supported by the positive regression slope.
The p-value proves significance.	The p-value indicates that the observed result would be unlikely under the null model.
The experiment was accurate.	The result was close to the accepted value, but precision and systematic error must be considered separately.

Banned phrase

“Human error” is not an evaluation. It is a placeholder. Replace it with the specific mechanism: parallax, endpoint overshoot, temperature drift, evaporation, contamination, calibration drift, biological variation, inconsistent timing, or incomplete mixing.

6 How to report p-values without inversion

A p-value should not be written as the probability that the hypothesis is true or false. It is conditional on the null hypothesis and the model assumptions.

Do not write	Write
There is a 3% chance that the null hypothesis is true.	If the null hypothesis were true, a result this extreme would be expected about 3% of the time under the test model.
The result is not due to chance.	The result would be unlikely under the null model.
The treatment definitely worked because $p < 0.05$.	The data provide statistical evidence for a treatment effect; effect size and limitations must also be considered.
Since $p > 0.05$, there is no difference.	No statistically significant difference was detected; the data may still be compatible with effects of practical importance.

Safe p-value sentence

A test gave $p = \dots$, meaning that data this extreme would be uncommon if the null hypothesis and the test assumptions were true.

7 Non-significant results

Non-significant results are often written badly. A high p-value does not prove that nothing is happening. It may reflect a small sample, high variability, weak measurement precision, or a

genuinely small effect.

Poor version

There was no difference between the two enzymes because $p = 0.21$.

Better version

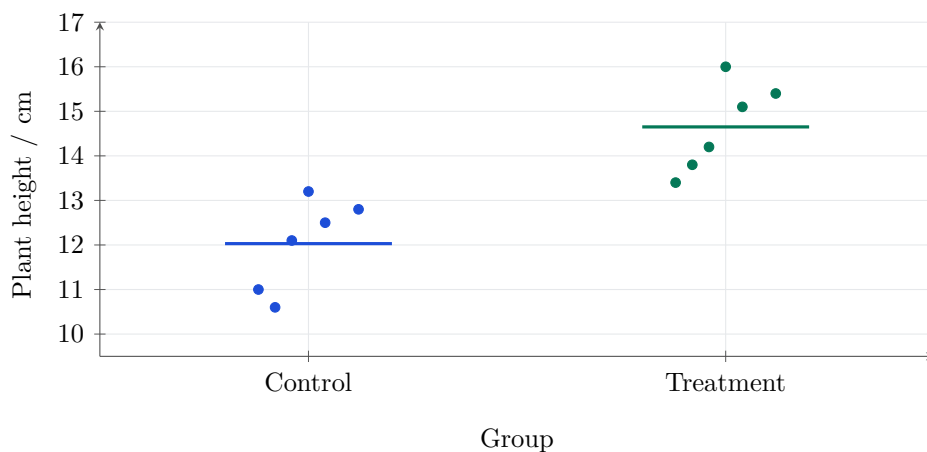
No statistically significant difference in mean activity was detected between the two enzymes, $p = 0.21$. However, the confidence interval was wide, so the data are still compatible with a moderate difference. A larger sample or more precise activity assay would be needed to make a stronger claim about equivalence.

Key distinction

Failure to reject H_0 is not the same as proving H_0 . To argue that two treatments are practically equivalent, use an equivalence framework or define a smallest meaningful effect in advance.

8 Graphs and claims

A graph should not decorate the conclusion. It should carry evidence. The text should point to the relevant visual features: direction, spread, overlap, outliers, curvature, or residual pattern.



Short horizontal lines show group means. The graph should be discussed in the conclusion, not merely inserted.

Graph-linked wording

The treatment group had a higher mean height, and the raw-data plot shows limited overlap between groups. The difference appears large relative to the within-group spread, but the sample size is small, so uncertainty should be reported alongside the mean difference.

9 Effect size and scientific meaning

A statistically significant result may be scientifically trivial. A non-significant result may still contain a large but uncertain observed effect. Therefore, write the effect size in scientific units whenever possible.

Situation	Effect size to report	Scientific wording
Two means	Mean difference, Cohen's d , or both	The treatment mean was 2.8 cm higher.
Regression	Slope and possibly R^2	Absorbance increased by 0.42 units per mol L ⁻¹ .
Survival data	Risk difference, relative risk, odds ratio	Mortality was 2.1 times higher in the exposed group.
Chi-square goodness-of-fit	Observed versus expected proportions, Cohen's w	The recessive phenotype was overrepresented relative to 3:1 expectation.
ANOVA	η^2 or ω^2	Treatment group explained about 31% of the variation in height.

Best habit

Report the result in the units of the investigation first. Standardized measures are useful, but students and readers need the real biological or chemical size of the effect.

10 Limitations as mechanisms

IB evaluation is often weak because students list generic errors. A good limitation explains a mechanism and its likely direction or consequence.

Weak limitation	Mechanistic version	Consequence for claim
Human error	Endpoint colour change was judged visually and may have been overshoot.	Titration volumes may be systematically too high.
Small sample size	Only six biological replicates were used per treatment.	The mean difference is uncertain and outliers have strong influence.
Temperature was not controlled	Enzyme tubes were handled sequentially, so later tubes may have warmed longer.	Apparent treatment effects may partly reflect temperature drift.
Contamination	Petri dishes were opened near the bench edge during plating.	Colony counts may include non-target colonies.
Instrument error	The colorimeter was zeroed once at the beginning but not checked between trials.	Absorbance readings may drift over time.

Quick rewrite

Rewrite this limitation:

"The results may be wrong because of human error."

Your version must name the mechanism, say how it affects the data, and explain how it limits the conclusion.

11 Templates for common IB analyses

These templates are not scripts to copy blindly. They are structures. Students should replace the placeholders with their own data, units, and limitations.

Two-group comparison

The [treatment] group had a higher/lower mean [response] than the [control] group. The observed mean difference was [value and unit]. A [test name] gave $p =$ [value], suggesting that the observed difference [would/would not] be unusual under the null hypothesis of equal means. This supports/does not strongly support the claim that [scientific claim], although [specific limitation] limits the strength of the conclusion.

Regression or calibration

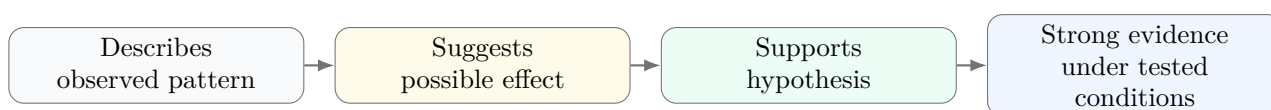
The scatter plot suggested a [positive/negative/curved] relationship between $[x]$ and $[y]$. The fitted model had slope [value and units], meaning that [scientific interpretation]. The residual plot [did/did not] show a strong pattern, so the linear model was [reasonable/questionable] within the tested range. The model should not be extrapolated beyond [range] because [reason].

Chi-square test

The observed counts [did/did not] closely match the expected counts under [expected ratio or independence model]. A chi-square test gave $p =$ [value], indicating that a deviation this large [would/would not] be unusual if the null model were true. The largest contribution came from [category/cell], suggesting that [interpretation]. However, [small expected count/design limitation] limits the conclusion.

12 Claim strength ladder

Scientific writing has degrees of strength. Students should choose the strength that the evidence deserves.



Stronger wording requires stronger design, clearer effect size, and lower uncertainty.

Avoid absolute wording

In school-level experimental work, avoid “proved”, “confirmed”, “definitely”, “no effect”, and “completely accurate”. These words usually overstate what the design can support.

13 Worked revision

Weak student version

The hypothesis was proven because the fertilizer plants were taller and the p-value was less than 0.05. Some errors happened because of human error, but overall the experiment was successful.

Improved version

The fertilizer group had a higher mean plant height than the control group. The observed mean difference was 2.8 cm, which is large relative to the within-group spread shown in the raw-data plot. A Welch t-test gave $p = 0.018$, so this difference would be unlikely under the null hypothesis of equal means. The data therefore support the claim that fertilizer increased plant height under the tested conditions. However, the conclusion is limited by the small number of biological replicates and possible light variation between trays, which could also affect growth.

Revision task

Improve the following conclusion:

“There was no relationship because the R^2 value was low. This was probably due to errors in measuring absorbance.”

Your version should mention the scatter plot, slope, residuals, possible measurement mechanism, and claim strength.

14 Final checklist

Before submitting a result or evaluation section, students should check the following.

Question	Evidence in the writing
Did I state the observed pattern?	Direction and group/variable names are clear.
Did I report the size of the effect?	Difference, slope, risk ratio, odds ratio, η^2 , or other relevant effect size.
Did I mention uncertainty?	SD, CI, overlap, sample size, or width of estimate.
Did I use the p-value correctly?	No inversion; no claim that p is the probability the hypothesis is true.
Did I avoid proving language?	Uses “suggests”, “supports”, or “is consistent with” when appropriate.
Did I connect statistics to science?	Biological or chemical meaning is explicit.
Did I evaluate mechanisms?	Limitations name specific sources of bias or uncertainty.
Did I restrict the claim?	The conclusion applies only to the tested range, organisms, chemicals, concentrations, and conditions.

Exit statement

Good statistical writing does not make the result sound stronger. It makes the claim more precise. A precise claim is more scientific than an exaggerated one.